

Possible Answers :

2722 to 2778

Question Number : 348 Question Id : 640653904379 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

Now, suppose BA realizes that the fixed costs per flight are in fact 41000 instead of 30000. It also finds out that two different types of passengers fly to Chennai. Type A consists of business people with a demand of $Q_A = 260 - 0.4P$ and type B consists of students with a demand of $Q_B = 240 - 0.6P$. BA decides to charge these two types different prices.

What is the total consumer surplus? $CS_{total} = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

21656 to 22094

Industry 4.0

Section Id :	64065364141
Section Number :	13
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	22
Number of Questions to be attempted :	22
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653134157
Question Shuffling Allowed :	No

Question Number : 349 Question Id : 640653904410 Question Type : MCQ Calculator : Yes

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533045289. ✓ YES

6406533045290. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653134158

Question Shuffling Allowed :

Yes

Question Number : 350 Question Id : 640653904411 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

Match Column A with Column B (country of origin)

Column A	Column B
A1. 1 st revolution	B1. England
A2. 2 nd revolution	B2. USA
A3. 3 rd revolution	B3. USA/Japan
A4. 4 th revolution	B4. Germany

Options :

6406533045291. ✓ A1:B1, A2:B2, A3:B3, A4:B4

6406533045292. ✗ A1:B2, A2:B1, A3:B3, A4:B4

6406533045293. ✗ A1:B3, A2:B1, A3:B4, A4:B2

6406533045294. ✗ A1:B3, A2:B1, A3: B2, A4:B4

Question Number : 351 Question Id : 640653904429 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets in the neighbouring shelves. He picked a few milk-powder packets, however these are not part of his list. He was searching for Good-day biscuits. He could identify only one packet in the shelf remaining. However, there were lot of Sun-fist biscuits lying in the next shelf. So he picked ten Sun- fist packets. Now map the scenarios with the items below:

Product	Scenario
P1)Tea:Coffee	S1)Complementary
P2) Tea: milk-powder packet	S2)Substitution
P3) Good-day biscuits: Sun-fist biscuit	S3)Inventory dependant demand
	S4)Impulse buying

Options :

6406533045332. ✖ P1-S1, P2-S3, P3-S4

6406533045333. ✔ P1-S2, P2-S1, P3-S3

6406533045334. ✖ P1-S2, P2-S1, P3-S4

Question Number : 352 Question Id : 640653904436 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following model corresponds to Exponential Smoothing forecasting model?

Options :

6406533045357. ✔ $F_{t+1} = D_{t+1} - F_t$ where F_t, D_t represents the forecast and actual data for period t

6406533045358. ✖ $F_t = \mu - \theta * \epsilon_{t-1}$ where $\epsilon_{t-1} = D_{t-1} - F_{t-1}$

Question Number : 353 Question Id : 640653904437 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

Select the correct form of internet from column B corresponding to the example in Column A.

Column A	Column B
A1. Smart metering	B1. Consumer
A2. Banking system	B2. Commercial
A3. Entertainment integrated devices	B3. Industrial
	B4. Private

Options :

6406533045359. ✖ A1: B3, A2: B2, A3:B4

6406533045360. ✔ A1: B3, A2: B2, A3:B1

6406533045361. ✖ A1: B2, A2: B3, A3:B1

Sub-Section Number :

3

Sub-Section Id :

640653134159

Question Shuffling Allowed :

Yes

Question Number : 354 Question Id : 640653904412 Question Type : MCQ Calculator : Yes

Correct Marks : 1.5

Question Label : Multiple Choice Question

Select the appropriate purposes of facility layout from the list below:

Options :

6406533045295. ✖ Maintain production flexibility

6406533045296. ✖ enhance sales as appropriate in manufacturing and service facilities.

6406533045297. ✔ Improve working capital

6406533045298. ✖ Promote high employee morale

Question Number : 355 Question Id : 640653904413 Question Type : MCQ Calculator : Yes

Correct Marks : 1.5

Question Label : Multiple Choice Question

Ashoke motors manufactures micro-cars in Pune plant. It sources various parts for a car from its different plants located across India and also from its subcontractors. The Pune plant assembles the final product. For assembling, different machines are involved. On 3rd of August, one of the machine caught fire and the entire assembly line stopped until a substitute machine arrived. What type of facility layout Ashoke Motors is maintaining at the Pune Plant?

Options :

6406533045299. ✔ Product layout

6406533045300. ✖ Process layout

6406533045301. ✖ Fixed position layout

Sub-Section Number :

4

Sub-Section Id :

640653134160

Question Shuffling Allowed :

Yes

Question Number : 356 Question Id : 640653904414 Question Type : SA Calculator : None

Correct Marks : 1.5

Question Label : Short Answer Question

A factory works for two 8-hour shifts each day and produces a total of 24,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds. How many workstations are required to main the specified level of production (assuming that activities can be combined into those workstations in any order) (enter your answer in numeric value)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number :

5

Sub-Section Id :

Question Shuffling Allowed :

640653134161
Yes

Question Number : 357

Question Id : 640653904415

Question Type : SA

Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

Calculate the Euclidean distance between two machines located at the coordinates A(5,10) and B(2,6) in a shop floor.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.8 to 5.3

Sub-Section Number :

Sub-Section Id :

Question Shuffling Allowed :

6

640653134162

Yes

Question Number : 358

Question Id : 640653904416

Question Type : MSQ

Calculator : Yes

Correct Marks : 1.5

Max. Selectable Options : 0

Question Label : Multiple Select Question

There are four demand points at A(1,2), B(2,3), C(3,5), D(4,1). The respective demand at these points are 7, 1, 3, 5. Select the optimal location based on cross-median approach.

Options :

6406533045304. ✓ (2,2)

6406533045305. ✓ (3,2)

6406533045306. ✓ (2.5,2)

6406533045307. ✗ (2,3)

Sub-Section Number :

Sub-Section Id :

Question Shuffling Allowed :

7

640653134163

Yes

Question Number : 359

Question Id : 640653904417

Question Type : MSQ

Calculator : Yes

Correct Marks : 2

Max. Selectable Options : 0

Question Label : Multiple Select Question

When is outsourcing not useful?

Options :

6406533045308. ✓ When the (production) volumes are large

6406533045309. ✓ Uncertainty in supply

6406533045310. ✓ If the service requires specific resources rather than service

6406533045311. ✓ Quality is impacted

Question Number : 360 Question Id : 640653904433 Question Type : MSQ Calculator : Yes
Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following supply chain costs is/are affected by distribution network structure?

Options :

6406533045345. ✓ Inventory

6406533045346. ✓ Transportation

6406533045347. ✓ Facilities and handling

6406533045348. ✓ Information

6406533045349. ✗ Price

Question Number : 361 Question Id : 640653904434 Question Type : MSQ Calculator : Yes
Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are considered in the Vehicle Routing Problem but not considered in TSP?

Options :

6406533045350. ✗ Subtour elimination

6406533045351. ✓ Any capacity restrictions

6406533045352. ✓ A specific starting point

6406533045353. ✗ Distance travelled restriction

Sub-Section Number :

8

Sub-Section Id :

640653134164

Question Shuffling Allowed :

No

Question Id : 640653904418 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Calculator : None

Question Numbers : (362 to 365)

Question Label : Comprehension

A contractor estimates that the size of the workforce needed over the next 5 weeks is 5, 7, 8, 4 and 6 workers, respectively. Excess labour kept on the force will cost \$300 per worker per week, and new hiring in any week will incur a fixed cost of \$400 plus @200 per worker per week. The objective is to minimize the total cost of labour needed for the project. If Dynamic Programming is applied to this problem, answer the given subquestions:

Sub questions

Question Number : 362 Question Id : 640653904419 Question Type : MCQ Calculator : Yes
Correct Marks : 1

Question Label : Multiple Choice Question

How is a stage represented by?

Options :

6406533045312. ✓ Week

6406533045313. ✗ Number of labours available in a week

6406533045314. ✗ Number of new hires in a week

6406533045315. ✗ Total cost of labour in a week.

Question Number : 363 Question Id : 640653904420 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

What are the alternatives in a stage?

Options :

6406533045316. ✗ Week

6406533045317. ✗ Number of new hires in a week

6406533045318. ✓ Number of labours in a week

6406533045319. ✗ Number of labours available at the end of previous week

6406533045320. ✗ Total cost of labour in a week.

Question Number : 364 Question Id : 640653904421 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

How state variable is defined for this problem?

Options :

6406533045321. ✗ Week

6406533045322. ✗ Number of labours in a week

6406533045323. ✓ Number of labours available at the end of previous week

6406533045324. ✗ Number of new hires in a week

6406533045325. ✗ Total cost of labour in a week.

Question Number : 365 Question Id : 640653904422 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

What is the optimal solution (objective function value) in Stage 5 when the state variable takes value 4?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

8

Sub-Section Number : 9
Sub-Section Id : 640653134165
Question Shuffling Allowed : No

Question Id : 640653904423 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (366 to 369)

Question Label : Comprehension

Bluescope company estimates its ordering cost at Rs 400 and carrying cost to be Rs.2/year/unit. The estimated annual requirement is 10,000 units at a unit price of Rs. 10. Calculate the given subquestions:

Sub questions

Question Number : 366 Question Id : 640653904424 Question Type : SA Calculator : None Correct Marks : 1

Question Label : Short Answer Question

What is the economic order quantity (EOQ)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2000

Question Number : 367 Question Id : 640653904425 Question Type : SA Calculator : None Correct Marks : 1

Question Label : Short Answer Question

Calculate the number of orders per year.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 368 Question Id : 640653904426 Question Type : SA Calculator : None Correct Marks : 1.5

Question Label : Short Answer Question

Calculate the annual inventory carrying cost.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2000

Question Number : 369 Question Id : 640653904427 Question Type : SA Calculator : None

Correct Marks : 1.5

Question Label : Short Answer Question

While rechecked, Bluescope company finds that the order cost should have been 20% more than the currently estimated value. What should be the percentage increase in the EOQ in this case? Enter your answer without the '%' symbol.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

9.4 to 9.6

Sub-Section Number : 10

Sub-Section Id : 640653134166

Question Shuffling Allowed : Yes

Question Number : 370 Question Id : 640653904428 Question Type : SA Calculator : None

Correct Marks : 3

Question Label : Short Answer Question

Colony Shop is stocking heavy coats for next winter. Colony pays \$50 for coat and sells it for \$110. At the end of the winter season, Colony Shop offers the coats at \$55 each. The demand for coats during the winter season is more than 20 but less than or equal to 30, all with equal probabilities. Determine the optimal order quantity that will maximize revenue for Colony Shop. Consider demand to be continuous.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

29 to 33

Sub-Section Number : 11

Sub-Section Id :

640653134167

Question Shuffling Allowed :

Yes

Question Number : 371 Question Id : 640653904430 Question Type : MSQ Calculator : Yes

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following example(s) can be referred as seasonal pattern?

Options :

6406533045335. ✖ Economic recession

6406533045336. ✔ Increases Ceiling Fan sales during winters

6406533045337. ✔ Increased tourist footfall towards the end of the year

Sub-Section Number :

12

Sub-Section Id :

640653134168

Question Shuffling Allowed :

Yes

Question Number : 372 Question Id : 640653904431 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following functions, Logistics Management does not cover?

Options :

6406533045338. ✖ Supply Management

6406533045339. ✖ Materials Management

6406533045340. ✔ Customer Management

6406533045341. ✖ Distribution Management

Question Number : 373 Question Id : 640653904432 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

What is the decision variable X_{ij} denotes in a Vehicle Routing Problem (VRP)?

Options :

6406533045342. ✔ If the vehicle travels node j from node i (binary)

6406533045343. ✖ Number of vehicles travelled between node i and node j (integer)

6406533045344. ✖ Weigh on the arc (i,j) e.g. the distance (integer)

Question Number : 374 Question Id : 640653904435 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

You visit the car service centre to get your car washed. In the service centre, washing done using automated systems, e.g. the car is put on a moving belt, water is sprayed, drying is also done.

Which of the following term represents this scenario?

Options :

6406533045354. ✖ Digitization

6406533045355. ✔ Automation

6406533045356. ✖ Digital transformation

Sub-Section Number : 13

Sub-Section Id : 640653134169

Question Shuffling Allowed : No

Question Id : 640653904438 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (375 to 376)

Question Label : Comprehension

Suppose you need to accept reservation for a flight. There are total 100 seats in the aircraft. You have two classes of passengers - economy and business. One economy seat gets you 2,500 Rs of revenue while a business seat gets 7,500 Rs. From your experience, you know the demand for business travellers follows a uniform distribution in [10,40]. Then answer the given subquestions:

Sub questions

Question Number : 375 Question Id : 640653904439 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

What is the Booking limit?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 376 Question Id : 640653904440 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

What is the protection level?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

70